

Regulating relative prices: mandated discounts and price menus^{*}

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Abstract

Brazil's half-price law requires cinemas to charge eligible consumers half the full price, while leaving the firm free to set the full price. This paper studies how such relative price regulation affects a monopolist's price menu. When the mandated discount is greater than the firm's preferred discount, the firm raises the full price and lowers the eligible price relative to voluntary price discrimination. This response is stronger when more consumers are eligible for the discount. Firms may also offer other promotions, and if the promotional price is chosen independently, it remains unchanged, although its advertised discount increases. Under a linear demand function and positive common marginal cost, attendance can rise while welfare falls because losses from deterred full-price admissions can outweigh the gains from additional eligible admissions. Under a general demand function, demand curvature and monopoly markups determine the local welfare effect of tightening the mandate around the voluntary price ratio. When the mandated discount exceeds the voluntary discount, a cap delivering the same eligible price restores the voluntary full price and raises profit, consumer surplus, and total welfare relative to the mandate.

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1 Introduction

Price regulation is often analysed as a constraint on a single price, but firms usually choose prices as part of a broader menu. A regulation that fixes the ratio between two prices constrains their relationship while leaving the firm to choose their level and the other prices in its menu. Hence, the incidence of the regulation depends on how the firm adjusts its whole menu. In this paper, I study the pricing and welfare consequences of this form of relative price regulation.

An example of this type of regulation is Brazil's half-price law. It requires cinemas, and other firms in other cultural industries, to sell tickets to students and other eligible groups at exactly half the posted full price, while leaving cinemas free to choose that full price and to offer any other voluntary promotion. A related policy exists in Spain, where residents of the Canary Islands, Balearic Islands, Ceuta, and Melilla receive a 75% discount on airfares, while airlines choose the fare on which the discount is calculated (Calzada and Fageda 2012; Fageda et al. 2016, 2017).¹ In both cases, the benefit received by eligible consumers is defined relative to a price chosen by the firm.

Motivated by this class of policies, I build a monopoly model of relative price regulation.² When the mandated discount is more generous than the firm would voluntarily offer, the firm cannot retain the prices it would independently choose for each consumer group. It then raises the full price, partly offsetting the mandated discount, while the eligible price remains below its voluntary level. A larger share of eligible consumers strengthens the incentive to raise the full price. Consequently, consumers who pay the full price bear part of the adjustment due to the regulation.

I then extend the model by adding a voluntary promotional ticket.³ When the promotion serves a separate consumer group and the firm can choose its effective price independently, the regulation leaves the promotional price unchanged. As

¹Similar programs exist in other EU member states, such as Portugal, Italy, France, and the UK.

²The monopoly setting is motivated by the Brazilian cinema industry. Approximately 71% of Brazilian municipalities with an active cinema have a single cinema complex.

³Administrative data from the Brazilian National Film Agency show that the share of promotional tickets increased from 17.6% in 2018 to 24.2% in 2025.

the posted full price rises, however, the firm increases the advertised promotional discount to preserve that price. Thus, a larger advertised discount can arise without any reduction in the promotional price paid by consumers.

To analyse the welfare effects of this regulation, I compare the regulated menu with voluntary third-degree price discrimination. The lower price for the eligible group attracts additional consumers, while the higher full price deters others. The welfare effect depends on the surplus generated by the additional eligible admissions relative to the surplus lost from the deterred full price admissions. Thus, total attendance can rise while welfare falls. With a general demand function, I show how demand curvature and monopoly markups determine the local welfare and output effects of moving away from the firm's voluntary price ratio. Under linear demand and a common positive marginal cost for both groups, I characterise these effects over the range of prices in which both groups remain served. A mandate slightly more generous than the firm's voluntary discount can increase attendance while reducing welfare. Under certain conditions, making the discount sufficiently more generous reverses the welfare effect.

To isolate the cost of linking the two prices, I compare the ratio mandate with an eligible price cap set to deliver the same eligible price. When the mandated discount is more generous than the firm's voluntary discount, both policies give eligible consumers the same price. However, the cap removes the link between the eligible and full prices, allowing the firm to restore the full price to its voluntary level. As a consequence, consumers facing the full price pay less, while eligible consumers face no change. The firm also earns more because it can choose its preferred full price. Thus, the matched cap raises profit, consumer surplus, and total welfare relative to the ratio mandate.

Related literature. This paper is related to the literature on the regulation of a firm's price structure. Price caps and average revenue constraints leave firms with some flexibility over individual prices but can distort relative prices within the regulated basket (Bradley and Price 1988; Armstrong et al. 1994; Cowan 1997a,b). Armstrong and Vickers (1991) study price discrimination under an average price

constraint, while Cowan (2018) studies regulations that equalise profit margins or price-cost ratios across markets. In the present paper, I study a regulator that fixes the ratio between two consumer prices and allows the firm to choose their common level. This imposes a different constraint in the firm's decision problem, and its incidence depends on how the firm adjusts the rest of its price menu. I characterise that adjustment and identify conditions under which the regulation increases total demand while reducing welfare.

This paper also contributes to the welfare analysis of third-degree price discrimination (Schmalensee 1981; Varian 1985; Aguirre et al. 2010; Cowan 2012, 2016). This literature primarily compares voluntary price discrimination with uniform pricing. I instead consider a price ratio chosen by a regulator, with uniform pricing as a special case in which the ratio equals one and voluntary third-degree price discrimination as the benchmark ratio preferred by the firm. For a general class of demand functions, I show how demand curvature and monopoly markups determine the local output and welfare effects of moving away from the voluntary ratio. Under linear demand, I characterise the comparison for larger departures and show that welfare can vary non-monotonically with the generosity of the mandated discount.

The regulation studied in this paper also connects to policies and contractual restrictions that transmit distortions across linked prices. For example, Medicaid's pricing rules tie public drug payments to private sector prices, while Spanish airfare subsidies base reimbursement on firm-chosen fares (Scott Morton 1997; Duggan and Scott Morton 2006; Calzada and Fageda 2012; Fageda et al. 2016, 2017). Platform parity clauses similarly restrict a seller's prices across channels (Boik and Corts 2016; Wang and Wright 2020). Brazil's half-price law differs from these arrangements because it fixes an exact ratio between prices paid by consumer groups within the same channel, without government purchase or reimbursement.

Finally, the policy that motivates this paper relates to research on pricing and price discrimination in cultural markets. de Roos and McKenzie (2014) study voluntary cinema discounts, while Leslie (2004) and Courty and Pagliero (2012) examine price discrimination in theatre and concerts. Davis (2006) and Einav (2007) study spatial

pricing and demand in film exhibition. For Brazil, [Wink Junior et al. \(2016\)](#) shows how half-price laws increase students' consumption of cultural goods. This paper complements that work by studying how a statutory price ratio changes the firm's optimal price menu and the welfare consequences of that adjustment.

The rest of the paper is organised as follows. Section 2 describes Brazil's half-price law and the observed price menu in the cinema industry. Section 3 develops the model and derives the pricing, welfare, and policy comparison results. Section 4 concludes. Proofs are in the appendix.

2 Institution, data, and stylised facts

2.1 Brazil's half-price law

Brazil has a long history of regional laws establishing a minimum discount for cultural and sporting events, beginning in 1990. A national law standardising these rules was approved in 2013. Law No. 12,933/2013 guarantees students admission at half the ticket price effectively charged to the general public and extends the benefit to people with disabilities and, when necessary, their companions, as well as low-income people aged 15-29.⁴ People aged 60 or above have a separate entitlement under the Elderly Citizen Statute, which specifies a discount of at least 50%.⁵

The federal legislation followed exhibitors' concerns about the expansion of half-price ticket sales and fraudulent student identification. According to the National Federation of Film Exhibition Companies, the share of half-price tickets increased from around 40% of cinema admissions in 2001 to 70% in 2007.⁶ The federal law standardised the documentation used to establish eligibility and limited the statutory entitlement to 40% of the tickets available for each event.⁷ Elderly consumers' separate entitlement does not contain this limit.⁸

⁴See Law No. 12,933/2013, Article 1, https://www.planalto.gov.br/ccivil_03/_ato2011-2014/2013/lei/l12933.htm.

⁵See Law No. 10,741/2003, Articles 1 and 23, https://www.planalto.gov.br/ccivil_03/leis/2003/l10.741.htm.

⁶See <https://www1.folha.uol.com.br/fsp/ilustrad/fq2904200707.htm> and <https://g1.globo.com/Noticias/Cinema/0,,MUL25098-7086,00.html>.

⁷Decree No. 8,537/2015 sets out the reservation and disclosure rules.

⁸See Law No. 12,933/2013, Article 1, paragraph 10, and Decree No. 8,537/2015, Articles 9–11, https://www.planalto.gov.br/ccivil_03/_ato2015-2018/2015/decree/d8537.htm.

Eligibility is based on legally defined characteristics, and the law explicitly determined the student identification cards used to establish eligibility. It recognises cards issued by the National Student Union, the National Student Union of Secondary School Students, and the National Association of Postgraduate Students.⁹ Other eligible groups establish their entitlement through documentation of age, disability, or qualifying low-income status.

Cinemas are still free to offer voluntary discounts that are not determined by the statutory ratio, and the statutory benefit is not cumulative with these promotions. For example, Cinemark offers discounted tickets through its partnership with Banco Bradesco, while UCI and Kinoplex have similar partnerships with Banco Itaú. These arrangements create a separate price tier whose price is chosen voluntarily by each cinema.

2.2 Data and the observed price menu

Each cinema in Brazil must report ticket sales for every film session to the Brazilian National Film Agency (ANCINE). Through an FOI request, I obtained yearly aggregates of revenue and admissions by ticket category for all Brazilian municipalities reporting cinema activity between 2018 and 2025.¹⁰

Table 1 summarises national aggregates for the three paid ticket categories: full-price, statutory half-price, and promotional tickets. Total admissions also include complimentary tickets, which account for approximately 1.5–2.5% of admissions over the period. Statutory half-price tickets are the largest category in every year, accounting for more than 50% of admissions. The full-price share falls from 26.5% in 2018 to 19.9% in 2025, while the promotional share rises from 17.6% to 24.2%. Thus, all three ticket categories are quantitatively relevant, and the decline in the full-price share is accompanied by an increase of the same magnitude in the promotional share.

[//www.planalto.gov.br/ccivil_03/_ato2015-2018/2015/decreto/d8537.htm](http://www.planalto.gov.br/ccivil_03/_ato2015-2018/2015/decreto/d8537.htm).

⁹Before the federal legislation, cinemas had to recognise cards issued by individual educational establishments.

¹⁰Legal restrictions prevent ANCINE from releasing the individual session records by ticket category. The category reporting requirement began in 2018, so comparable information is not available for earlier years.

Table 1. National aggregates, 2018–2025

Year	N municipalities	Tickets (M)	PMI (R\$)	Full-price (R\$)	Half-price (R\$)	Promo (R\$)	% Full-price	% Half-price	% Promo	Half/full	
2018		405	164.4	14.89	21.82	12.80	12.36	26.5	54.2	17.6	0.587
2019		422	177.6	15.61	24.61	13.72	12.54	21.5	59.4	17.4	0.558
2020		426	39.9	15.88	25.44	14.00	13.00	20.0	61.0	17.3	0.550
2021		411	51.7	17.20	26.81	14.34	13.15	26.1	58.4	14.1	0.535
2022		423	97.9	18.80	29.38	16.39	13.92	23.4	59.6	15.4	0.558
2023		432	118.1	19.35	29.91	17.57	13.87	22.0	59.6	16.7	0.587
2024		447	129.2	19.58	30.96	18.14	14.42	19.9	57.4	20.8	0.586
2025		454	117.2	20.00	31.41	19.12	14.63	19.9	53.5	24.2	0.609

Note: N municipalities counts municipalities; M denotes millions. Full-price, half-price, and promotional prices are computed by dividing category revenue by category admissions, in nominal BRL. Omitted complimentary tickets account for around 1.5%–2.5% over the time period. PMI is total box-office revenue divided by all admissions.

ANCINE reports total revenue and admissions within each category. Thus, I compute the average ticket price for each category by dividing its revenue by its admissions. Between 2018 and 2025, the full price increases by 44.0%, close to cumulative inflation of 45.2%. The average promotional price increases by only 18.3%, implying a real decline of approximately 18.5%. Average box-office revenue per admission, reported in the PMI column, increases by 34.3% and falls by 7.5% in real terms.¹¹ Total admissions fall from 164.4 million in 2018 to 118.1 million in 2023 and 117.2 million in 2025, indicating that the sector had not returned to its pre-pandemic attendance level by the end of the period.

2.3 Aggregation and market structure

The half-price/full-price ratio in Table 1 does not reproduce the statutory ratio exactly. Each category average price aggregates across cinemas, films, screening times, formats, and locations. Eligible and full price consumers may purchase different combinations of screenings even when every individual cinema complies with the law. Thus, the average price ratio cannot be interpreted as a direct measure of compliance. In a conversation with a director of a large cinema chain, they confirmed that compliance with the ratio is strict.

A separate ANCINE dataset containing session-level attendance provides information about whether the statutory quota is likely to constrain half-price sales. Average occupancy was approximately 21% over 2017-2019. The proportion of sessions with attendance above 40% of room capacity was 14.5% in 2018, 11.2% in 2024, and 9.3% in 2025. Hence, in most sessions, total attendance was below the statutory quota. In

¹¹The inflation adjustment uses annual IPCA changes published by the Brazilian Institute of Geography and Statistics.

addition, in a conversation with a director of a large cinema chain, they reported that, in the cinema industry, firms sell half-price tickets to eligible consumers even after discounted sales exceed 40% of session capacity.¹² Together, these facts suggest that every eligible consumer can purchase the statutory ticket for a cinema session.

Finally, Table 2 describes the structure of local cinema exhibition markets. Approximately 71% of municipalities with a cinema screen have a single active cinema complex.¹³

Table 2. Market structure by year

Year	Screens	Complexes	Municipalities	Single complex municipalities	Share (%)
2018	3281	784	396	282	71.2
2019	3447	822	414	292	70.5
2020	1825	446	272	202	74.3
2021	3238	748	402	290	72.1
2022	3378	793	419	300	71.6
2023	3434	823	430	305	70.9
2024	3499	856	443	317	71.6
2025	3516	868	447	317	70.9

3 Model

Relative price regulation fixes the price paid by eligible consumers as a fraction of the posted full price but leaves the firm free to choose the price level. This allows the firm to partly offset a mandated discount by changing the reference price. This section models that response and its implications for the rest of the price menu and for welfare.

The institutional details in Section 2.1 motivate the main features of the model. Consistent with the prevalence of municipalities served by a single cinema complex, a monopolist chooses the posted full price. Eligible consumers pay a regulated fraction of that price. I assume that any statutory quota is slack, so all eligible consumers

¹²This practice may differ in other cultural activities, such as music festivals, where organisers usually restrict discounted sales to the statutory quota.

¹³Table 1 includes every municipality in the revenue and admissions dataset, whereas Table 2 includes only municipalities matched to ANCINE's yearly cinema-complex data. The latter records the cinemas active in December of each year. A municipality in which a cinema closed before December may therefore appear in the revenue and admissions data but not in the cinema-complex dataset. Except for 2020, the coverage difference is small. Coverage is substantially lower in the pandemic year because many cinemas were not active in December.

can purchase the good. The firm may also offer a promotional ticket whose price it chooses voluntarily.

I first characterise how the regulated ratio affects the posted full price and then introduce the promotional ticket. Next, I compare welfare under the regulation with welfare under voluntary third-degree price discrimination and under a cap that delivers the same eligible price. The final extensions allow promotional prices to be fixed percentages of the posted price and statutory and promotional eligibility to overlap.

3.1 Setup and benchmark

There is a unit mass of consumers. A mass $\alpha_H \in (0, 1)$ is eligible for the regulated discount, while the remaining $1 - \alpha_H$ is not. The subscripts F and H index the non-eligible and eligible groups, respectively. Group membership is observable and fixed, and consumers cannot purchase at another group's price.

Let $\varphi_j(p_j)$ denote the firm's per-capita operating profit from charging group $j \in \{F, H\}$ the price p_j . I assume that the relevant price range is an interval and that each φ_j is twice continuously differentiable, satisfies $\varphi_j''(p_j) < 0$ on that interval, and has a unique interior maximiser.

Strict concavity guarantees a unique optimum, and the negative second derivatives ensure that the price comparative statics are well defined. I also assume that the profit function is separable across groups, which allows each group's contribution to profit to be evaluated independently. For example, with a common positive marginal cost c , per-capita profit can take the form $\varphi_j(p_j) = (p_j - c)q_j(p_j)$.¹⁴

The firm chooses the posted full price p . Non-eligible consumers pay $p_F = p$, while eligible consumers pay

$$p_H = r p, \quad r \in (0, 1], \quad (1)$$

¹⁴The pricing results also permit group-specific costs or fees, provided that the resulting per-capita profit functions satisfy the same curvature and separability assumptions.

where r is fixed by the regulator. For instance, in the Brazilian cinema case, this corresponds to $r = 1/2$. Under this regulated ratio, profit is

$$\pi_R(p) = (1 - \alpha_H)\varphi_F(p) + \alpha_H\varphi_H(rp). \quad (2)$$

Let $p_R^*(r, \alpha_H)$ denote the firm's optimal posted price under the regulated ratio r and eligible share α_H . When doing so causes no ambiguity, I suppress the population share argument and write the optimal posted price as $p_R^*(r)$.

I compare the regulation against voluntary third-degree price discrimination, under which the firm chooses the two group prices independently. For each group j , define $p_j^{PD} \equiv \arg \max_{p_j} \varphi_j(p_j) > 0$. This gives the unconstrained prices p_F^{PD} and p_H^{PD} . The firm's preferred price ratio is

$$r^* \equiv \frac{p_H^{PD}}{p_F^{PD}}. \quad (3)$$

The regulation binds whenever $r \neq r^*$. If $r < r^*$, the mandated discount is more generous than the firm would voluntarily offer; if $r > r^*$, it is less generous. When $r = r^*$, the regulation reproduces the voluntary price ratio.

3.2 Relative price regulation and the posted full price

The two terms in equation (2) have different preferred posted prices. The non-eligible group's profit is maximised at p_F^{PD} , while the eligible group's profit is maximised at the posted price p_H^{PD}/r , which implements its voluntary price. The firm chooses a common posted price to maximise their weighted sum. When $r = 1$, this is the usual uniform-pricing problem.

PROPOSITION 1 (Posted price under relative price regulation). *Suppose there is an interval of feasible posted prices containing p_F^{PD} and p_H^{PD}/r on which both $\varphi_F(p)$ and $\varphi_H(rp)$ are defined. Then:*

- (i) If $r < r^*$, the mandated discount is more generous than the firm's preferred discount.
The full price rises and the eligible price falls relative to voluntary discrimination:

$$p_F^{PD} < p_R^*(r, \alpha_H) < \frac{p_H^{PD}}{r}, \quad \frac{\partial p_R^*(r, \alpha_H)}{\partial \alpha_H} > 0.$$

- (ii) If $r > r^*$, the mandated discount is less generous than the firm's preferred discount.
The full price falls and the eligible price rises relative to voluntary discrimination:

$$\frac{p_H^{PD}}{r} < p_R^*(r, \alpha_H) < p_F^{PD}, \quad \frac{\partial p_R^*(r, \alpha_H)}{\partial \alpha_H} < 0.$$

- (iii) If $r = r^*$, the regulation reproduces the voluntary prices and

$$p_R^*(r, \alpha_H) = p_F^{PD} = \frac{p_H^{PD}}{r}, \quad \frac{\partial p_R^*(r, \alpha_H)}{\partial \alpha_H} = 0.$$

When $r < r^*$, the firm cannot retain both voluntary prices. At the voluntary full price, the mandated eligible price is below the eligible group's profit-maximising level. Raising the posted price moves the eligible price towards that optimum but moves the non-eligible price away from its own optimum. The firm therefore chooses a compromise in which non-eligible consumers pay more and eligible consumers pay less than under voluntary discrimination.

A larger eligible share puts more weight on the profit coming from the eligible group, so the firm raises the posted price further. When $r > r^*$, the relative positions of the two voluntary benchmarks are reversed, so both price and eligible share comparative statics are reversed.

3.3 Relative price regulation and promotional discounts

As described in Section 2.1, the firm may also offer a voluntary promotional ticket alongside the regulated ticket. For example, the promotional group can represent consumers who have access to a bank card partnership. I incorporate this additional price by introducing a third consumer group. A mass α_P has access to a promotional ticket, a mass α_H is eligible for the regulated price, and the remaining mass $\alpha_F = 1 - \alpha_H - \alpha_P$ pays the full price.

I assume, initially, that the three groups are mutually exclusive and have strictly positive masses. I extend the notation above so that φ_j , for every $j \in \{F, H, P\}$, is twice continuously differentiable, satisfies $\varphi_j'' < 0$, and has a unique interior maximiser p_j^{PD} .

The firm chooses the posted full price p and the promotional discount $d \in [0, 1)$. The resulting price menu is

$$p_F = p, \quad p_H = r p, \quad p_P = p(1 - d).$$

For an interior promotional discount, choosing (p, d) is equivalent to choosing the posted and effective promotional prices (p, p_P) subject to $0 < p_P < p$. Thus, I can write the profit as

$$\pi_R(p, p_P) = \alpha_F \varphi_F(p) + \alpha_H \varphi_H(r p) + \alpha_P \varphi_P(p_P). \quad (4)$$

When the promotional price constraint is slack, p_P enters as a separate term in the objective function. The first two terms determine the posted price, while the third determines the promotional price. Thus, only the relative sizes of the non-eligible and eligible groups matter for the posted-price choice. Define the eligible share among these two groups as

$$\theta \equiv \frac{\alpha_H}{\alpha_F + \alpha_H} = \frac{\alpha_H}{1 - \alpha_P}. \quad (5)$$

Apart from the positive factor $1 - \alpha_P$, the part of profit that depends on p is

$$(1 - \theta)\varphi_F(p) + \theta\varphi_H(rp),$$

which is the problem in Proposition 1 with θ as the eligible share.

To compare this menu with third-degree price discrimination, define the voluntary promotional discount relative to the full price as

$$d_V^* \equiv 1 - \frac{p_P^{PD}}{p_F^{PD}}.$$

Under regulation, let $p_R^*(r, \theta)$ denote the optimal posted price and let

$$d_R^* \equiv 1 - \frac{p_P^{PD}}{p_R^*(r, \theta)}.$$

Applying Proposition 1 with θ in place of α_H yields the following corollary.

COROLLARY 1 (Promotional price separation benchmark). *Suppose there is an interval of feasible posted prices containing p_F^{PD} and p_H^{PD}/r on which both $\varphi_F(p)$ and $\varphi_H(rp)$ are defined. Suppose also that the voluntary and regulated promotional discounts are interior, so that $p_F^{PD} > p_P^{PD}$ and $p_R^*(r, \theta) > p_P^{PD}$. Then:*

(i) *The effective promotional price is*

$$p_P^R = p_P^{PD},$$

independently of the regulated ratio and the population shares.

(ii) If $r < r^*$, relative price regulation raises the posted price and deepens the promotional discount:

$$p_F^{PD} < p_R^*(r, \theta) < \frac{p_H^{PD}}{r}, \quad d_R^* > d_V^*,$$

and

$$\frac{\partial p_R^*(r, \theta)}{\partial \theta} > 0, \quad \frac{\partial d_R^*}{\partial \theta} > 0.$$

(iii) If $r > r^*$,

$$\frac{p_H^{PD}}{r} < p_R^*(r, \theta) < p_F^{PD}, \quad d_R^* < d_V^*,$$

and both comparative statics are negative. If $r = r^*$, the regulated and voluntary menus coincide and both derivatives are zero.

This corollary shows that the promotional tier does not change the effective price charged to promotional consumers. The firm uses the discount d to keep that price at p_P^{PD} . Regulation does, however, change the discount expressed relative to the posted price. When regulation raises the posted price, the firm must advertise a larger percentage discount to preserve the same promotional price. Thus, a larger advertised markdown does not necessarily imply a lower transaction price.

The posted full price depends on the population shares through θ . Holding α_P fixed,

$$\frac{\partial \theta}{\partial \alpha_H} = \frac{1}{1 - \alpha_P} > 0,$$

so the comparative statics with respect to α_H have the same signs as those in the corollary.

These results rely on profit separability, fixed group membership, and an interior promotional discount. If the constraint $p_P \leq p$ binds, the firm cannot necessarily retain p_P^{PD} . Section 3.6 considers fixed-percentage promotions and overlapping statutory and promotional eligibility.

3.4 Welfare effects of the ratio constraint

I now analyse the welfare effects of relative price regulation. When $r < r^*$, Proposition 1 shows that the regulation raises the posted full price while lowering the eligible price relative to voluntary discrimination. The lower eligible price attracts additional eligible consumers and transfers surplus to existing eligible consumers. The higher posted price, however, deters some non-eligible consumers from purchasing the good. The welfare effect depends on the surplus gained from additional eligible consumers relative to the surplus lost from deterred non-eligible consumers.

As before, I compare the ratio mandate with voluntary third-degree price discrimination under the promotional price separation benchmark of Corollary 1. Promotional consumers pay p_P^{PD} in both regimes, so their welfare contribution cancels. Hence, I only need to compare the non-eligible and eligible groups while allowing a positive promotional share.

I define total welfare as profit plus consumer surplus, with marginal costs measuring resource costs and equal welfare weights across consumers. Thus, transfers from the firm to existing consumers do not change total welfare. I first consider linear demand with a common marginal cost, then examine a local result under general demand.

3.4.1 Welfare under linear demand

Consider first a numerical example with no promotional group. Let the two groups have equal mass, demands $q_F(p) = 18 - p$ and $q_H(p) = 10 - p$, and common marginal cost $c = 2$. The voluntary prices are $(p_F^{PD}, p_H^{PD}) = (10, 6)$. A half-price mandate changes the price vector to $(10.4, 5.2)$.

In this example, admissions by non-eligible consumers fall by 0.2, while eligible

admissions rise by 0.4, so total attendance rises by 0.2. The lost admissions by non-eligible consumers reduce surplus by 1.64, while the additional eligible admissions add 1.44, reducing total welfare by 0.20. Higher attendance does not raise welfare because the lost admissions by non-eligible consumers generate more surplus per admission than the additional eligible admissions.

To formally characterise this trade-off, suppose each group has linear per-capita demand $q_j(p_j) = a_j - b_j p_j$, where $a_j, b_j > 0$, and a common marginal cost $c \geq 0$ satisfying $c < a_j/b_j$. I restrict attention to ratios for which both groups purchase positive quantities at the regulated optimum:

$$p_R^*(r) < \frac{a_F}{b_F}, \quad r p_R^*(r) < \frac{a_H}{b_H}.$$

The voluntary monopoly prices are $m_j \equiv p_j^{PD} = (a_j + b_j c)/(2b_j)$. Define

$$A \equiv \alpha_F b_F, \quad B \equiv \alpha_H b_H, \quad D \equiv A + B r^2, \quad z \equiv m_H - r m_F.$$

The weights A and B combine each group's mass and demand slope. The gap z measures how far the voluntary eligible price lies above the price required by the mandate at the voluntary full price. Thus, $z > 0$ exactly when $r < r^*$. Let $W_R(r)$ and $Q_R(r)$ denote total welfare and output under the mandate, and W_V and Q_V their counterparts under voluntary discrimination.

PROPOSITION 2 (Attendance and welfare under linear demand). *Under the promotional price separation benchmark, linear demand, common marginal cost $c \geq 0$, and interior solutions, the optimal posted price is*

$$p_R^*(r) = \frac{A m_F + B r m_H}{D}.$$

The changes in total output and welfare relative to voluntary discrimination are

$$Q_R(r) - Q_V = \frac{AB(1-r)}{D} z, \tag{6}$$

$$W_R(r) - W_V = \frac{AB}{2D} z [z - 2c(1-r)]. \tag{7}$$

For $0 < r < \min\{r^*, 1\}$, the mandate raises total output. It lowers welfare if and only if

$$0 < z < 2c(1 - r).$$

With zero marginal cost, welfare is instead weakly higher than under voluntary discrimination, with equality only when $z = 0$.

The output result reflects the relative quantities gained and lost. When $r < r^*$, eligible admissions rise by ABz/D , while non-eligible admissions fall by $ABr z/D$. For $r < 1$, the gain is larger than the loss. Welfare, however, weights each admission by willingness to pay less marginal cost. The lost non-eligible admissions can generate more surplus per admission than the additional eligible admissions, so the increase in attendance need not raise welfare.

With positive common marginal cost and $r^* < 1$, a mandate slightly more generous than the voluntary discount lowers welfare: z is small and positive, so the welfare-loss condition holds. If $m_H > 2c$, a sufficiently more generous mandate reverses this comparison, provided both groups remain served. Appendix A.2 derives the ratio threshold and the welfare-minimising ratio. For a half-price mandate, the welfare-loss condition is particularly simple:

$$0 < m_H - \frac{1}{2}m_F < c.$$

The numerical example satisfies this condition because $m_H - m_F/2 = 1 < c = 2$.

A geometric interpretation. Figure 1 illustrates why the cost assumption matters. With linear demand, profit falls with the squared distance from the voluntary price menu $m = (m_F, m_H)$, while welfare falls with the squared distance from the efficient menu (c, c) . These distances weight the two prices by A and B . The firm therefore chooses the point on the regulated ray $p_H = r p_F$ closest to m , but that point can lie farther from the efficient menu. This is the welfare-loss region shown in the figure. Appendix A.2 gives the formal projection argument and extends the welfare formula to group-specific costs.

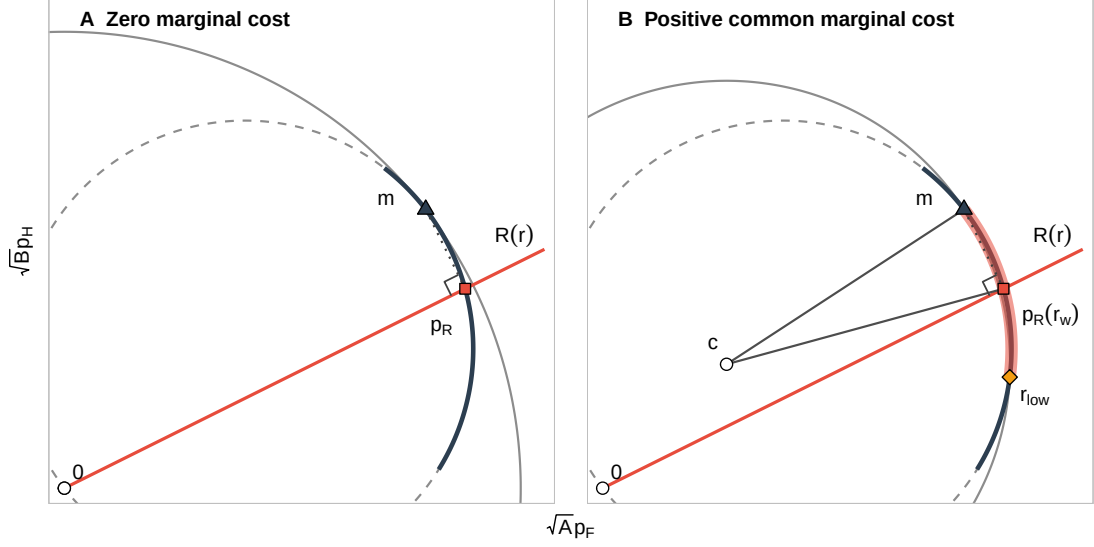


Figure 1. The geometry of relative price regulation in weighted price coordinates. The axes scale group prices by the square roots of $A = \alpha_F b_F$ and $B = \alpha_H b_H$. The vector m contains the voluntary monopoly prices, and the cost vector has both entries equal to c . The dashed curve is the locus of regulated optima, sometimes called the Thales circle, with diameter $[0, m]$; its dark arc is traced by the regulated menu $(p_R^*(r), r p_R^*(r))$ over the feasible ratios shown. The solid grey circle is the welfare contour through m . At zero cost, the feasible arc lies inside this contour. With positive common cost, the red portion lies outside it and represents the welfare-loss region.

The model also nests uniform pricing. At $r = 1$, total output is unchanged and welfare rises by $AB(m_H - m_F)^2/[2(A + B)]$, recovering the familiar linear-demand comparison with voluntary third-degree price discrimination (Schmalensee 1981; Varian 1985).

3.4.2 A local result under general demand

The increase in attendance accompanied by a welfare loss can also arise under general demand. With common positive marginal cost, a voluntary ratio below one, and equal curvature indices across groups, a small reduction in the ratio raises output but lowers welfare. More generally, the local comparison depends on both demand curvature and monopoly markups. The result evaluates a departure from the voluntary ratio; the effects of larger departures can differ as prices, markups, and curvature change.

For $j \in \{F, H\}$, let $q_j(p_j)$ be positive, twice continuously differentiable, and strictly

decreasing near the voluntary monopoly price $m_j \equiv p_j^{PD}$. With constant marginal cost $c_j < m_j$, per-capita profit is $\varphi_j(p_j) = (p_j - c_j)q_j(p_j)$. As before, the voluntary ratio is $r^* = m_H/m_F$.

Holding group shares fixed, define the Lerner index L_j and curvature index χ_j as

$$L_j \equiv \frac{m_j - c_j}{m_j}, \quad \chi_j \equiv -\frac{(m_j - c_j)\varphi_j''(m_j)}{q_j(m_j)} = 2 - \frac{(m_j - c_j)^2 q_j''(m_j)}{q_j(m_j)}.$$

The index L_j measures the markup as a share of price. The index χ_j measures how quickly marginal profit falls around the voluntary optimum, normalised by the markup and sales; it equals two under linear demand. Let $CS_R(r)$ and CS_V denote consumer surplus under the mandate and voluntary discrimination, respectively.

PROPOSITION 3 (General demand near the voluntary ratio). *Under the conditions above, suppose $r^* \in (0, 1]$. A sufficiently small reduction in the regulated ratio below r^* :*

- (i) *raises welfare and consumer surplus if $\chi_H/L_H < \chi_F/L_F$, and lowers both if the inequality is reversed;*
- (ii) *raises total output if $\chi_H < \chi_F/r^*$, and lowers it if the inequality is reversed.*

At r^ , the welfare and consumer-surplus derivatives with respect to r coincide (using left derivatives if $r^* = 1$). If either comparison holds with equality, the corresponding first-order effect is zero and higher-order terms determine the effect of a reduction in the ratio.*

Proposition 3 shows that demand curvature and markups jointly determine the local welfare effect. With a common positive marginal cost, $r^* < 1$ implies $m_H < m_F$ and therefore $L_H < L_F$. If the two groups have equal curvature indices, lowering the ratio below r^* then lowers welfare. At the same time, part (ii) implies that the same reduction raises output.

Under linear demand, $\chi_F = \chi_H = 2$, so the local welfare comparison depends on the two Lerner indices alone. The general result therefore reproduces the local implication of Proposition 2.

With zero marginal cost, $L_F = L_H = 1$, and only the curvature comparison $\chi_H - \chi_F$

remains. A larger normalised convexity term

$$\frac{m_H^2 q_H''(m_H)}{q_H(m_H)}$$

reduces χ_H and favours a welfare-improving reduction in the regulated ratio, echoing the curvature conditions in the price-discrimination literature (Aguirre et al. 2010; Cowan 2016). Under zero-cost linear demand, the first-order welfare effect is zero; the positive welfare gain in Proposition 2 is second order in the departure from the voluntary ratio.

The equality of the welfare and consumer surplus derivatives follows because r^* implements the voluntary profit-maximising price vector. Maximised profit is therefore locally stationary with respect to r at the voluntary ratio. The proof gives the full derivative expressions.

3.5 Relative price regulation and an eligible price cap

To separate the effect of the discount from the effect of linking the two prices, I compare the ratio mandate with a cap fixing the price paid by eligible consumers. I set the cap equal to the eligible price produced by the ratio mandate:

$$\bar{p}_H = r p_R^*(r).$$

When this ceiling binds, the two instruments deliver the same eligible price. This matched comparison holds the eligible price fixed to isolate the effect of linking prices. The matching cap is market-specific: its level depends on the demand, costs, and consumer composition that determine the outcome under the mandate.

The comparison is particularly simple when $r < r^*$. Under the ratio mandate, the full price is above its voluntary optimum. Under the cap, the firm can hold the eligible price fixed while choosing the full and promotional prices independently. Hence, it

restores those prices to their voluntary optima. Eligible and promotional consumers face the same prices under the two instruments, while non-eligible consumers pay less and the firm earns more under the cap.

Let the superscripts V , R , and C denote voluntary discrimination, relative price regulation, and the matched cap. Let $W_C(r)$ denote welfare under the cap. Under the cap, the firm chooses p_F , p_H , and p_P separately, subject only to

$$p_H \leq \bar{p}_H.$$

As before, I omit the promotional group's welfare contribution because the regulation does not affect its final price.

PROPOSITION 4 (Relative price regulation versus an eligible price cap). *Suppose that the assumptions of Corollary 1 hold, consumer surplus for each group is strictly decreasing in its price over the relevant range, and total welfare is the sum of profit and group consumer surplus. Then:*

(i) *If $r < r^*$, the cap binds and produces*

$$(p_F^C, p_H^C, p_P^C) = (p_F^{PD}, r p_R^*(r), p_P^{PD}).$$

Relative-price regulation produces

$$(p_F^R, p_H^R, p_P^R) = (p_R^*(r), r p_R^*(r), p_P^{PD}), \quad p_R^*(r) > p_F^{PD}.$$

The cap yields strictly higher profit, consumer surplus, and total welfare than relative-price regulation. It also restores the voluntary promotional discount:

$$d_C^* = d_V^* < d_R^*.$$

(ii) *If $r = r^*$, the cap, relative price regulation, and voluntary discrimination produce the same price menu.*

(iii) If $r > r^*$, the matched cap is slack and produces the voluntary price menu. It cannot reproduce the higher eligible price under relative price regulation, and the general profit function assumptions do not determine the welfare ranking.

Intuitively, both instruments give eligible consumers the same price, but the cap removes the link between that price and the rest of the menu. Non-eligible consumers and the firm are strictly better off, while eligible and promotional consumers face unchanged prices. The cap's dominance therefore survives any additional non-negative welfare weight placed on eligible consumers.

Decomposing welfare. The cap comparison can also decompose the welfare effect of the ratio mandate. For any r , define the decoupled price menu

$$(p_F^D, p_H^D, p_P^D) = (p_F^{PD}, r p_R^*(r), p_P^{PD})$$

and let $W_D(r)$ denote its welfare. This menu combines the eligible price produced by the ratio mandate with the voluntary prices for the other groups. When $r < r^*$, it is exactly the outcome of the matched cap, so

$$W_D(r) = W_C(r).$$

For every r , the decoupled menu separates the welfare effect of changing the eligible price from the cost of distorting the rest of the menu:

$$W_R(r) - W_V = \underbrace{W_D(r) - W_V}_{\text{eligible price component}} - \underbrace{W_D(r) - W_R(r)}_{\text{menu distortion}}. \quad (8)$$

Under general downward-sloping demand and constant group-specific marginal costs, the two components are

$$W_D(r) - W_V = \alpha_H \int_{r p_R^*(r)}^{m_H} (p - c_H) |q'_H(p)| dp, \quad (9)$$

$$W_D(r) - W_R(r) = \alpha_F \int_{m_F}^{p_R^*(r)} (p - c_F) |q'_F(p)| dp. \quad (10)$$

For $r < r^*$, the first component measures the surplus generated by the additional eligible admissions. It is positive if the eligible price remains above marginal cost. If the eligible price falls below marginal cost, the mandate can instead induce admissions whose resource cost exceeds willingness to pay. The second component measures the surplus lost when the higher full price deters consumers. The welfare effect of the ratio mandate is the first component less the second.

Under linear demand with $z > 0$, the welfare loss occurs when r times the non-eligible group's average markup exceeds the eligible group's average markup. With common positive marginal cost and $r^* < 1$, the voluntary eligible price has the smaller price-cost margin. Additional eligible admissions can therefore contribute less surplus per ticket than the admissions lost among non-eligible consumers. Welfare falls when the surplus lost on deterred non-eligible admissions exceeds the surplus gained on additional eligible admissions. The proof gives the linear-demand expressions for these two components.

When $r > r^*$, the cap is slack and no longer reproduces the mandate's eligible price. Its outcome is the voluntary menu, so either welfare ranking is possible. [Appendix A.5](#) gives examples of both rankings.

The form of the entitlement. The two instruments also differ in the form of the entitlement. When $r < r^*$, the cap produces the realised price ratio

$$\frac{p_H^C}{p_F^C} = \frac{r p_R^*(r)}{p_F^{PD}} > r.$$

Eligible consumers pay the same absolute price under both instruments, but the

displayed percentage discount is smaller under the cap. Thus, a cap can reproduce the eligible price without reproducing a legal entitlement expressed as “half price.” The ratio makes the percentage entitlement directly observable while creating the menu distortion identified above.

3.6 Extensions to promotional pricing

The independent promotional price benchmark allows the firm to adjust the promotional discount separately in every market. I now consider two alternatives. First, the firm may fix the promotional percentage across markets, causing the effective promotional price to move with the posted price. Second, statutory and promotional eligibility may overlap, so that some consumers choose the lower of the two discounted prices.

3.6.1 Fixed-percentage promotions

Let $\rho \equiv 1 - d \in (0, 1)$ and suppose that a chain fixes ρ across its locations during a given period. The promotional price is then $p_P = \rho p$, and the posted-price problem becomes

$$\pi_\rho(p) = \alpha_F \varphi_F(p) + \alpha_H \varphi_H(\rho p) + \alpha_P \varphi_P(\rho p). \quad (11)$$

The promotional term is now linked to the posted price, so the firm balances three profit components when choosing that price. The optimal posted price must lie between the three transformed group-specific optima. The following proposition formalises this trade-off and its comparative statics.

PROPOSITION 5 (Fixed-percentage promotions). *Suppose the three profit functions in equation (11) satisfy the maintained negative second derivative assumption, their transformed maximisers lie in the feasible interval, and all three group shares are positive. Then π_ρ has a unique maximiser p_ρ^* lying between*

$$p_F^{PD}, \quad \frac{p_H^{PD}}{r}, \quad \frac{p_P^{PD}}{\rho}.$$

For any market characteristic v that changes the optimal posted price but leaves ρ fixed,

$$\frac{\partial \log p_P}{\partial v} = \frac{\partial \log p_\rho^*}{\partial v}.$$

Holding α_P fixed and increasing α_H at the expense of α_F gives

$$\frac{\partial p_\rho^*}{\partial \alpha_H} = - \frac{r \phi'_H(r p_\rho^*) - \phi'_F(p_\rho^*)}{\pi''_\rho(p_\rho^*)}.$$

Increasing α_P at the expense of α_F gives the analogous expression with

$$\rho \phi'_P(\rho p_\rho^*)$$

in place of

$$r \phi'_H(r p_\rho^*).$$

In particular, if

$$p_F^{PD} \leq \frac{p_P^{PD}}{\rho} \leq \frac{p_H^{PD}}{r} \quad \text{and} \quad p_F^{PD} < \frac{p_H^{PD}}{r},$$

then

$$\frac{\partial p_\rho^*}{\partial \alpha_H} > 0.$$

A fixed-percentage promotional price is a third ratio-linked term in the transformed

common-price problem. Because $p_P = \rho p$, the promotional price moves one-for-one in logs with the posted price. In this setting, the promotional group also affects the optimal posted price.

Under the ordering in the proposition, the promotional benchmark lies between the non-eligible group's voluntary benchmark and the posted price that would implement the eligible group's voluntary optimum. This ordering preserves the positive comparative static of the posted price with respect to the eligible share. It does not, by itself, determine the effect of the promotional share, which depends on the relative marginal profits of the promotional and non-eligible groups.

The observed promotional category can therefore combine two forms of pricing. Some promotional prices may follow the separation benchmark and remain independent of the posted price. Others may be set as a fixed percentage of the posted price and move with it.

3.6.2 Overlapping eligibility

The separation benchmark also assumes that statutory and promotional eligibility are disjoint. In practice, a consumer may qualify for both discounts and pay

$$\min\{r p, p_P\}.$$

Three regimes are possible. If $p_P < r p$, consumers eligible for both discounts use the promotion. If $r p < p_P$, they use the statutory discount. When $p_P = r p$, the two discounts coincide and the firm's objective has a kink.

Let us use the partition of the population into four mutually exclusive groups with masses

$$\alpha_F, \quad \alpha_{H \setminus P}, \quad \alpha_{P \setminus H}, \quad \alpha_B,$$

which sum to one. The groups $H \setminus P$ and $P \setminus H$ contain consumers eligible only for the statutory discount and only for the promotion. Group B contains consumers eligible for both.

I focus on the kink

$$p_P = r p.$$

Immediately below the kink, changing p_P affects both promotional-only and dual-eligible consumers. Immediately above it, changing p_P affects only promotional-only consumers. These two marginal effects determine whether the firm chooses coincident promotional and statutory prices.

Let φ_B denote per-capita profit from a dual-eligible consumer. At a given posted price p , setting $p_P = r p$ is locally optimal with respect to p_P when

$$\alpha_{P \setminus H} \varphi'_P(r p) + \alpha_B \varphi'_B(r p) \geq 0$$

and

$$\alpha_{P \setminus H} \varphi'_P(r p) \leq 0.$$

Define the pooled promotional optimum

$$p_{P \cup B}^* \equiv \arg \max_x \{ \alpha_{P \setminus H} \varphi_P(x) + \alpha_B \varphi_B(x) \}.$$

Strict concavity makes the two inequalities equivalent to

$$p_P^{PD} \leq r p \leq p_{P \cup B}^*.$$

Conditional on the posted price, the promotional price coincides with the statutory price when the mandated eligible price lies between the promotional-only optimum and the pooled optimum for promotional-only and dual-eligible consumers. A joint optimum at the kink must also support the posted-price choice analysed below. This interval is non-empty only when pooling the dual-eligible group weakly raises the preferred promotional price.

When the kink binds, the part of profit that depends on the posted price is

$$\alpha_F \varphi_F(p) + \alpha_{H \setminus P} \varphi_H(rp) + \alpha_{P \setminus H} \varphi_P(rp) + \alpha_B \varphi_B(rp).$$

Define the total mass whose price is linked to the statutory ratio as

$$\beta \equiv \alpha_{H \setminus P} + \alpha_{P \setminus H} + \alpha_B$$

and define its average per-capita profit function as

$$\bar{\varphi}_E(x) \equiv \frac{\alpha_{H \setminus P} \varphi_H(x) + \alpha_{P \setminus H} \varphi_P(x) + \alpha_B \varphi_B(x)}{\beta}.$$

The posted price-dependent part of profit can then be written as

$$\alpha_F \varphi_F(p) + \beta \bar{\varphi}_E(rp).$$

Holding the composition of the ratio-linked group fixed, Proposition 1 applies with β in place of α_H and $\bar{\varphi}_E$ in place of φ_H . Within a regime in which the kink remains optimal, overlapping eligibility therefore expands the mass whose price is tied to the posted price. An increase in that mass raises the posted price when the pooled group's preferred ratio exceeds r .

Together, the separation and overlapping-eligibility cases provide two benchmarks. When the promotional price can be chosen independently, regulation changes the advertised markdown but not the promotional transaction price. When the promotion coincides with the statutory price, the promotional tier becomes another ratio-linked component of the menu.

4 Conclusion

This paper studies relative price regulation, which fixes the price paid by eligible consumers as a fraction of the posted full price paid by non-eligible consumers, whilst allowing the firm to choose the price level and any other prices in its menu.

I show that when the regulation requires a larger percentage discount than the firm would voluntarily offer, the firm raises the posted full price and lowers the eligible price compared to voluntary discrimination. This response becomes stronger as the share of eligible consumers increases, shifting more of the cost of the regulation onto consumers who pay the full price. Additionally, if the firm can offer an independently chosen promotional price, that price remains unchanged even though the higher full price makes the advertised percentage discount larger.

The welfare effect of this regulation depends on how consumption changes across the two groups. The lower regulated price increases consumption among eligible consumers, but the higher full price reduces consumption among non-eligible ones. Thus, total output can rise whilst total welfare falls if the purchases deterred among full-price consumers would have generated more surplus than the additional purchases by eligible consumers. With a linear demand function and a common positive marginal cost, I derive the exact condition for this result. For example, with a half-price mandate, and provided that both groups remain served, welfare falls when

the voluntary eligible price is above half the voluntary full price, but the difference is smaller than marginal cost. With a general demand function, the local result also depends on demand curvature and the monopoly markups in the two groups.

To separate the effect of the discount from the effect of linking the two prices, I compare the ratio mandate with a cap fixing the price paid by eligible consumers. The cap removes the firm's incentive to raise the posted full price, returning it to its voluntary level, while the eligible price remains unchanged. Hence, non-eligible consumers pay less, eligible consumers pay the same price, and the firm earns a higher profit. Consumer surplus and total welfare are also greater under the cap. This comparison points to the central trade-off of the regulation. It leaves the price level to the firm, but doing so lets the firm shift part of the adjustment onto the unregulated reference price.

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Appendix A. Proofs and additional results

Proof of Proposition 1. For this proof, write $p_R^* = p_R^*(r, \alpha_H)$. The first-order condition for the regulated problem is

$$G(p, \alpha_H) \equiv (1 - \alpha_H)\phi'_F(p) + \alpha_H r \phi'_H(r p) = 0.$$

Because

$$G_p(p, \alpha_H) = (1 - \alpha_H)\phi''_F(p) + \alpha_H r^2 \phi''_H(r p) < 0,$$

the regulated objective is strictly concave, so any interior solution is unique.

At the voluntary full price,

$$G(p_F^{PD}, \alpha_H) = \alpha_H r \phi'_H(r p_F^{PD}),$$

because $\phi'_F(p_F^{PD}) = 0$. Strict concavity of ϕ_H implies

$$\text{sign}\left[\phi'_H(r p_F^{PD})\right] = \text{sign}(p_H^{PD} - r p_F^{PD}) = \text{sign}(r^* - r).$$

At the price that implements the eligible group's voluntary price,

$$G\left(\frac{p_H^{PD}}{r}, \alpha_H\right) = (1 - \alpha_H)\phi'_F\left(\frac{p_H^{PD}}{r}\right),$$

because $\phi'_H(p_H^{PD}) = 0$.

If $r < r^*$, then $p_F^{PD} < p_H^{PD}/r$. Consequently,

$$G(p_F^{PD}, \alpha_H) > 0 \quad \text{and} \quad G\left(\frac{p_H^{PD}}{r}, \alpha_H\right) < 0.$$

Continuity gives a zero between the two endpoints, and because G is strictly decreas-

ing in p , that zero is unique. It satisfies

$$p_F^{PD} < p_R^* < \frac{p_H^{PD}}{r}.$$

If $r > r^*$, the endpoint inequalities and the signs of G are reversed, giving

$$\frac{p_H^{PD}}{r} < p_R^* < p_F^{PD}.$$

If $r = r^*$, the two endpoints coincide and $p_R^* = p_F^{PD} = p_H^{PD}/r$.

To obtain the comparative static, differentiate the first-order condition implicitly:

$$\frac{\partial p_R^*}{\partial \alpha_H} = -\frac{G_{\alpha_H}}{G_p}.$$

At the optimum,

$$G_{\alpha_H} = -\varphi'_F(p_R^*) + r\varphi'_H(rp_R^*) = -\frac{1}{\alpha_H}\varphi'_F(p_R^*),$$

where the second equality uses the first-order condition. Therefore,

$$\frac{\partial p_R^*}{\partial \alpha_H} = \frac{\varphi'_F(p_R^*)}{\alpha_H [(1 - \alpha_H)\varphi''_F(p_R^*) + \alpha_H r^2 \varphi''_H(rp_R^*)]}.$$

The denominator is negative. Strict concavity of φ_F implies that $\varphi'_F(p_R^*) < 0$ when $p_R^* > p_F^{PD}$ and $\varphi'_F(p_R^*) > 0$ when $p_R^* < p_F^{PD}$. Hence

$$\text{sign}\left(\frac{\partial p_R^*}{\partial \alpha_H}\right) = \text{sign}(p_R^* - p_F^{PD}) = \text{sign}(r^* - r).$$

□

Proof of Corollary 1. For this proof, write $p_R^* = p_R^*(r, \theta)$. For an interior promotional discount, choosing (p, d) is equivalent to choosing (p, p_P) , where $p_P = p(1 - d)$, subject to $0 < p_P < p$. By equation (4), profit is

$$\pi_R(p, p_P) = \alpha_F \varphi_F(p) + \alpha_H \varphi_H(rp) + \alpha_P \varphi_P(p_P).$$

Because the objective is separable in p and p_p , and φ_p has the unique interior maximiser p_p^{PD} , the optimal effective promotional price is

$$p_p^R = p_p^{PD}.$$

The maintained condition $p_R^* > p_p^{PD}$ ensures that the constraint $p_p < p$ is slack at the regulated solution. Similarly, $p_F^{PD} > p_p^{PD}$ ensures an interior promotional discount under voluntary discrimination. The effective promotional price is therefore independent of the regulated ratio and the population shares.

Using equation (5), the part of profit that depends on the posted price can be written as

$$(1 - \alpha_p) [(1 - \theta)\varphi_F(p) + \theta\varphi_H(rp)].$$

The positive factor $1 - \alpha_p$ does not affect the optimum. Proposition 1 therefore applies directly with θ in place of α_H . Hence

$$\text{sign}(p_R^* - p_F^{PD}) = \text{sign}\left(\frac{\partial p_R^*}{\partial \theta}\right) = \text{sign}(r^* - r),$$

with p_R^* lying strictly between p_F^{PD} and p_H^{PD}/r whenever $r \neq r^*$.

Finally,

$$d_R^* - d_V^* = p_p^{PD} \left(\frac{1}{p_F^{PD}} - \frac{1}{p_R^*} \right) = \frac{p_p^{PD}(p_R^* - p_F^{PD})}{p_F^{PD} p_R^*},$$

so the discount comparison has the same sign as the posted-price comparison. Moreover,

$$\frac{\partial d_R^*}{\partial \theta} = \frac{p_p^{PD}}{(p_R^*)^2} \frac{\partial p_R^*}{\partial \theta}.$$

Thus the discount comparative static also has the sign of $r^* - r$. When $r = r^*$, the posted prices and discounts coincide and both derivatives are zero. $\square$

A.1 Linear-demand pricing formulas

For closed-form pricing formulas, specialise to linear per-capita demand and zero marginal cost:

$$q_j(p_j) = a_j - b_j p_j, \quad a_j > 0, \quad b_j > 0.$$

The parameters (a_j, b_j) describe per-capita demand and do not incorporate group size. In this case, $\varphi_j(p_j) = p_j q_j(p_j)$, and voluntary discrimination gives

$$p_F^{PD} = \frac{a_F}{2b_F}, \quad p_H^{PD} = \frac{a_H}{2b_H}, \quad (\text{A.1})$$

so that

$$r^* = \frac{a_H b_F}{a_F b_H}.$$

The optimal regulated posted price in the two-group model is

$$p_R^*(r, \alpha_H) = \frac{(1 - \alpha_H)a_F + r\alpha_H a_H}{2[(1 - \alpha_H)b_F + r^2\alpha_H b_H]}. \quad (\text{A.2})$$

An interior solution requires $p_R^*(r, \alpha_H) < a_F/b_F$ and $r p_R^*(r, \alpha_H) < a_H/b_H$. At $r = 1/2$, equation (A.2) becomes

$$p_{HL}^*(\alpha_H) = \frac{2(1 - \alpha_H)a_F + \alpha_H a_H}{4(1 - \alpha_H)b_F + \alpha_H b_H}. \quad (\text{A.3})$$

If the two groups have identical per-capita demand, so that $a_F = a_H = a$ and $b_F = b_H = b$, then

$$p_{HL}^*(\alpha_H) = \frac{a(2 - \alpha_H)}{b(4 - 3\alpha_H)}, \quad \frac{\partial p_{HL}^*}{\partial \alpha_H} = \frac{2a}{b(4 - 3\alpha_H)^2} > 0.$$

Taking the log derivative gives

$$\frac{\partial \log p_{HL}^*}{\partial \alpha_H} = \frac{2}{(4 - 3\alpha_H)(2 - \alpha_H)}, \quad (\text{A.4})$$

which is positive and increasing in $\alpha_H \in (0, 1)$.

With the promotional group included, the linear model gives

$$p_P^{PD} = \frac{a_P}{2b_P}, \quad (\text{A.5})$$

and

$$p_R^*(r, \theta) = \frac{\alpha_F a_F + r \alpha_H a_H}{2(\alpha_F b_F + r^2 \alpha_H b_H)}, \quad d_R^* = 1 - \frac{p_P^{PD}}{p_R^*(r, \theta)}. \quad (\text{A.6})$$

These formulas illustrate the separation benchmark in Corollary 1.

A.2 Linear-demand welfare: geometry and additional results

Proof of Proposition 2. I prove the result allowing constant group-specific marginal costs $c_j < a_j/b_j$, and then specialise to $c_F = c_H = c$. Write $m_j = (a_j + b_j c_j)/(2b_j)$ and retain the definitions $A = \alpha_F b_F$, $B = \alpha_H b_H$, $D = A + Br^2$, and $z = m_H - r m_F$. For vectors x and y , define

$$\langle x, y \rangle_{A,B} = Ax_F y_F + Bx_H y_H, \quad \|x\|_{A,B}^2 = \langle x, x \rangle_{A,B}.$$

Let $m = (m_F, m_H)$, $\mathbf{c} = (c_F, c_H)$, and $\mathbf{p} = (p_F, p_H)$. The regulated ray is $\mathcal{R}(r) = \{(p_F, p_H) : p_H = r p_F, p_F \geq 0\}$, and the regulated menu is $\mathbf{p}_R^*(r) = (p_R^*(r), r p_R^*(r))$.

Under the separation benchmark, group P is charged p_P^{PD} in both regimes, so its contribution to welfare cancels. For groups F and H , linear demand with constant marginal cost implies that per-capita profit can be written as

$$\varphi_j(p_j) = \varphi_j(m_j) - b_j(p_j - m_j)^2.$$

Thus aggregate profit is a constant minus $\|\mathbf{p} - m\|_{A,B}^2$. Minimising this squared distance over $\mathcal{R}(r)$ gives

$$p_R^*(r) = \frac{Am_F + Brm_H}{D}.$$

Equivalently, define the residual $u \equiv m - \mathbf{p}_R^*(r)$ by

$$u_F \equiv m_F - p_R^*(r) = -\frac{Brz}{D}, \quad u_H \equiv m_H - r p_R^*(r) = \frac{Az}{D}.$$

It satisfies

$$\langle u, (1, r) \rangle_{A,B} = Au_F + Bru_H = 0,$$

so $\mathbf{p}_R^*(r)$ is the weighted orthogonal projection of m onto the regulated ray.

Linear-demand total welfare can be written as a constant minus $\|\mathbf{p} - \mathbf{c}\|_{A,B}^2/2$. Hence

$$W_R(r) - W_V = \frac{1}{2} \left(\|m - \mathbf{c}\|_{A,B}^2 - \|\mathbf{p}_R^*(r) - \mathbf{c}\|_{A,B}^2 \right).$$

Expanding this difference and using the orthogonality condition gives

$$\begin{aligned} W_R(r) - W_V &= \frac{1}{2} \left(Au_F^2 + Bu_H^2 \right) + Au_F(p_R^*(r) - c_F) + Bu_H(r p_R^*(r) - c_H) \\ &= \frac{1}{2} \left(Au_F^2 + Bu_H^2 \right) + p_R^*(r)(Au_F + Bru_H) - (Ac_F u_F + Bc_H u_H) \\ &= \frac{1}{2} \left(Au_F^2 + Bu_H^2 \right) - (Ac_F u_F + Bc_H u_H) \\ &= \frac{AB}{2D} z^2 - \frac{AB}{D} z(c_H - rc_F) \\ &= \frac{AB}{2D} z \left[z - 2(c_H - rc_F) \right], \end{aligned}$$

Thus the projection representation and the general-cost welfare formula are

$$W_R(r) - W_V = \frac{1}{2} \|u\|_{A,B}^2 - \langle u, \mathbf{c} \rangle_{A,B}, \tag{A.7}$$

$$= \frac{AB}{2D} z \left[z - 2(c_H - rc_F) \right]. \tag{A.8}$$

The first term in equation (A.7) is the zero-cost welfare gain; the second is the marginal cost correction. Since u is orthogonal to the regulated ray, only the component of the cost vector orthogonal to that ray enters the correction. Setting $c_F = c_H = c$ gives equation (7).

For total output, linear demand gives

$$\begin{aligned}
Q_R(r) - Q_V &= A(m_F - p_R^*(r)) + B(m_H - r p_R^*(r)) \\
&= Au_F + Bu_H \\
&= \frac{AB(1-r)}{D} z.
\end{aligned}$$

This proves equation (6). For arbitrary c_F and c_H ,

$$W_R(r) < W_V \iff z[z - 2(c_H - rc_F)] < 0.$$

The inequality holds exactly when z lies strictly between the two roots 0 and $2(c_H - rc_F)$.

Holding r and the cost vector fixed gives a complementary characterisation in voluntary price space. The welfare-loss region is the strip, restricted to the maintained interior domain, between

$$m_H - rm_F = 0$$

and

$$m_H - rm_F = 2(c_H - rc_F).$$

The first boundary contains voluntary price menus that already satisfy the mandate. On the second boundary, the two components of the welfare change exactly offset.

If $c_F = c_H = 0$, the expression reduces to $ABz^2/(2D)$, which is non-negative and equals zero if and only if $z = 0$. If $c_F = c_H = c > 0$ and $0 < r < r^* < 1$, then $z > 0$ and $c_H - rc_F = c(1-r) > 0$. The welfare difference is negative if and only if $0 < z < 2c(1-r)$, and equation (6) is strictly positive throughout this interval.

When $m_F > 2c$, substituting $z = m_H - rm_F$ into the upper inequality gives

$$m_H - rm_F < 2c(1-r) \iff r > \frac{m_H - 2c}{m_F - 2c} \equiv r_{\text{low}}.$$

If $m_H > 2c$, then $0 < r_{\text{low}} < r^*$, which establishes the non-monotonicity result. If $m_H \leq 2c$, define $g(r) \equiv 2c(1-r) - z = 2c - m_H + r(m_F - 2c)$. This affine function is weakly positive at both endpoints of $[0, r^*]$: $g(0) = 2c - m_H \geq 0$ and $g(r^*) = 2c(1 - r^*) > 0$. Hence it is positive for every $r \in (0, r^*)$, apart from a possible equality at the excluded endpoint $r = 0$. Finally, setting $r = 1/2$ in $0 < z < 2c(1 - r)$ gives $0 < m_H - m_F/2 < c$.

For completeness, consider how welfare varies with the ratio when $c > 0$, $r^* < 1$, and $m_H > 2c$. The projection condition implies that the regulated menus trace an arc of the circle with diameter joining the origin and the voluntary menu in weighted coordinates. Welfare contours are circles centred at the cost vector. To locate the welfare minimum, write

$$\tilde{m} = (\sqrt{A}m_F, \sqrt{B}m_H), \quad \tilde{c} = (\sqrt{A}c, \sqrt{B}c).$$

The Thales circle has centre $\tilde{m}/2$ and radius $\|\tilde{m}\|/2$. Its point farthest from $\tilde{c}$ is

$$\frac{\tilde{m}}{2} + \frac{\|\tilde{m}\|}{2} \frac{\tilde{m} - 2\tilde{c}}{\|\tilde{m} - 2\tilde{c}\|}.$$

When $m_H > 2c$ and $r^* < 1$, both coordinates of $\tilde{m} - 2\tilde{c}$ are positive, so this point belongs to the arc defined by ratios in $(0, 1)$. Its interpretation as a welfare minimum also requires positive demand from both groups at the corresponding prices. Define

$$\lambda \equiv \frac{\sqrt{Am_F^2 + Bm_H^2}}{\sqrt{A(m_F - 2c)^2 + B(m_H - 2c)^2}}.$$

Transforming back to price coordinates gives

$$r_w = \frac{m_H + \lambda(m_H - 2c)}{m_F + \lambda(m_F - 2c)}.$$

This ratio lies strictly between $r_{\text{low}} = (m_H - 2c)/(m_F - 2c)$ and $r^* = m_H/m_F$ by the strict mediant inequality. Squared distance from $\tilde{c}$ increases along the feasible arc up to this point and decreases thereafter. Since welfare is a constant minus one-half of that squared distance, it falls up to r_w and then rises. On any closed interval throughout which the interior solution applies, welfare is therefore maximised at an endpoint.

Extending this comparison to all of $[0, 1]$ requires the interior conditions to hold throughout, including at the limiting case $r = 0$. $\square$

A.3 General-demand welfare

Proof of Proposition 3. Define $\gamma_j = -\varphi_j''(m_j) > 0$ and $K = \alpha_F \gamma_F + \alpha_H (r^*)^2 \gamma_H > 0$. If $r^* = 1$, all derivatives at the voluntary ratio are left derivatives.

At $r = r^*$, the constrained and voluntary price vectors coincide: $p_R^*(r^*) = m_F$ and $r^* p_R^*(r^*) = m_H$. Differentiating the firm's first-order condition

$$\alpha_F \varphi_F'(p_R^*(r)) + \alpha_H r \varphi_H'(r p_R^*(r)) = 0$$

with respect to r and evaluating at r^* gives

$$p_R^{*'}(r^*) = -\frac{\alpha_H r^* m_F \gamma_H}{K}, \quad \left. \frac{d(r p_R^*(r))}{dr} \right|_{r=r^*} = \frac{\alpha_F m_F \gamma_F}{K}.$$

Group j 's total-surplus derivative with respect to its price is $(p_j - c_j)q_j'(p_j)$. The monopoly first-order condition implies

$$q_j(m_j) + (m_j - c_j)q_j'(m_j) = 0,$$

so $W_j'(m_j) = -q_j(m_j)$. Applying the chain rule and substituting the two price derivatives therefore yields

$$\begin{aligned} \left. \frac{dW_R}{dr} \right|_{r=r^*} &= -\alpha_F q_F(m_F) p_R^{*'}(r^*) - \alpha_H q_H(m_H) \left. \frac{d(r p_R^*(r))}{dr} \right|_{r=r^*} \\ &= \frac{\alpha_F \alpha_H m_F}{K} [r^* \gamma_H q_F(m_F) - \gamma_F q_H(m_H)] \\ &= \frac{\alpha_F \alpha_H q_F(m_F) q_H(m_H)}{K} \left(\frac{\chi_H}{L_H} - \frac{\chi_F}{L_F} \right). \end{aligned}$$

The last equality uses $m_H = r^* m_F$ and $\gamma_j = \chi_j q_j(m_j)/(L_j m_j)$.

Let $\pi_R^*(r)$ denote maximised profit under the ratio. The envelope theorem gives

$$\left. \frac{d\pi_R^*}{dr} \right|_{r=r^*} = \alpha_H m_F \varphi_H'(m_H) = 0.$$

Since $W_R(r) = CS_R(r) + \pi_R^*(r)$, the welfare and consumer-surplus derivatives coincide at r^* . Consequently,

$$\left. \frac{dW_R}{dr} \right|_{r=r^*} = \left. \frac{dCS_R}{dr} \right|_{r=r^*} = \frac{\alpha_F \alpha_H q_F(m_F) q_H(m_H)}{K} \left(\frac{\chi_H}{L_H} - \frac{\chi_F}{L_F} \right). \quad (\text{A.9})$$

The same steps for total output give

$$\begin{aligned} \left. \frac{dQ_R}{dr} \right|_{r=r^*} &= \alpha_F q'_F(m_F) p_R^*(r^*) + \alpha_H q'_H(m_H) \left. \frac{d(r p_R^*(r))}{dr} \right|_{r=r^*} \\ &= \frac{\alpha_F \alpha_H q_F(m_F) q_H(m_H)}{m_F K L_F L_H} \left(\chi_H - \frac{\chi_F}{r^*} \right). \end{aligned}$$

Thus

$$\left. \frac{dQ_R}{dr} \right|_{r=r^*} = \frac{\alpha_F \alpha_H q_F(m_F) q_H(m_H)}{m_F K L_F L_H} \left(\chi_H - \frac{\chi_F}{r^*} \right). \quad (\text{A.10})$$

All prefactors are positive. Because the regimes coincide at r^* , a sufficiently small reduction in r changes welfare, consumer surplus, and output in the directions opposite to the signs of their respective derivatives. A zero derivative leaves the comparison to higher-order terms. $\square$

A.4 The matched cap and welfare decomposition

Proof of Proposition 4. Under the eligible-price cap, profit is

$$\pi_C(p_F, p_H, p_P) = \alpha_F \varphi_F(p_F) + \alpha_H \varphi_H(p_H) + \alpha_P \varphi_P(p_P), \quad p_H \leq \bar{p}_H.$$

The objective is separable. The unique unconstrained maximisers are therefore p_F^{PD} , p_H^{PD} , and p_P^{PD} .

For any r , define the decoupled menu

$$(p_F^D, p_H^D, p_P^D) = (p_F^{PD}, r p_R^*(r), p_P^{PD}).$$

Adding and subtracting its welfare gives

$$W_R(r) - W_V = (W_D(r) - W_V) - (W_D(r) - W_R(r)).$$

Since group j 's total-surplus derivative is $(p - c_j)q'_j(p)$, and the two comparisons change only the eligible and full prices, respectively,

$$W_D(r) - W_V = \alpha_H \int_{r p_R^*(r)}^{m_H} (p - c_H) |q'_H(p)| dp,$$

$$W_D(r) - W_R(r) = \alpha_F \int_{m_F}^{p_R^*(r)} (p - c_F) |q'_F(p)| dp,$$

which proves equations (9) and (10) for every r . Under linear demand, $m_H - r p_R^*(r) = Az/D$ and $p_R^*(r) - m_F = Brz/D$. Evaluating the integrals gives the following expressions.

$$W_D(r) - W_V = \frac{ABz}{2D} [m_H + r p_R^*(r) - 2c_H], \quad (\text{A.11})$$

$$W_D(r) - W_R(r) = \frac{ABr z}{2D} [m_F + p_R^*(r) - 2c_F]. \quad (\text{A.12})$$

Each component equals the change in the relevant group's aggregate admissions multiplied by its average price-cost margin over the adjustment. Subtracting the second component from the first reproduces equation (A.8).

Suppose first that $r < r^*$. Corollary 1 implies

$$p_F^{PD} < p_R^*(r) < \frac{p_H^{PD}}{r}.$$

Consequently, $\bar{p}_H = r p_R^*(r) < p_H^{PD}$, so the cap binds. The optimal menu under the cap is

$$(p_F^C, p_H^C, p_P^C) = (p_F^{PD}, r p_R^*(r), p_P^{PD}).$$

Under relative-price regulation, the menu is

$$(p_F^R, p_H^R, p_P^R) = (p_R^*(r), r p_R^*(r), p_P^{PD}).$$

The two regimes give the eligible and promotional groups identical prices. For the full-price group, strict concavity and the uniqueness of p_F^{PD} give

$$\pi_C - \pi_R = \alpha_F \left[\varphi_F(p_F^{PD}) - \varphi_F(p_R^*(r)) \right] > 0.$$

Because $p_F^{PD} < p_R^*(r)$ and full-price consumer surplus is strictly decreasing in price, aggregate consumer surplus is also strictly higher under the cap. Total welfare, defined as the sum of profit and consumer surplus, is therefore strictly higher.

Moreover,

$$d_C^* = 1 - \frac{p_P^{PD}}{p_F^{PD}} = d_V^*, \quad d_R^* = 1 - \frac{p_P^{PD}}{p_R^*(r)}.$$

Since $p_R^*(r) > p_F^{PD}$, it follows that $d_R^* > d_C^* = d_V^*$.

If $r = r^*$, Corollary 1 gives $p_R^*(r) = p_F^{PD}$ and $r p_R^*(r) = p_H^{PD}$. The cap is weakly binding and all three regimes produce the voluntary menu.

Finally, if $r > r^*$, Corollary 1 gives

$$\frac{p_H^{PD}}{r} < p_R^*(r) < p_F^{PD}.$$

Hence $\bar{p}_H = r p_R^*(r) > p_H^{PD}$, so the cap is slack and the firm chooses the voluntary menu $(p_F^{PD}, p_H^{PD}, p_P^{PD})$. The welfare comparison is therefore the comparison between relative-price regulation and voluntary discrimination. The two linear-demand examples in Appendix A.5 satisfy the maintained assumptions and produce opposite welfare rankings, establishing that the general assumptions do not determine the sign. $\square$

A.5 Welfare comparisons when the cap is slack

When $r > r^*$, the matched cap is slack and cannot reproduce the eligible price under the ratio mandate. Either welfare ranking is then possible. To illustrate, consider the two-group linear model with

$$\alpha_H = \frac{1}{2}, \quad b_F = b_H = 1, \quad r = 0.8.$$

First, let

$$(a_F, c_F) = (10, 0), \quad (a_H, c_H) = (6, 0).$$

Then $r^* = 0.6 < r$, the cap is slack, and

$$W_R(0.8) - W_C(0.8) = 0.152.$$

Alternatively, let

$$(a_F, c_F) = (12, 8), \quad (a_H, c_H) = (10, 0).$$

Then $r^* = 0.5 < r$, both regimes remain interior, and

$$W_R(0.8) - W_C(0.8) = -4.482.$$

Thus, when the cap is slack, relative price regulation can produce either higher or lower welfare than the cap. This ambiguity contrasts with the strict dominance of the cap when $r < r^*$.

A.6 Fixed-percentage promotions

Proof of Proposition 5. Define the transformed profit functions

$$\psi_H(p) \equiv \varphi_H(rp), \quad \psi_P(p) \equiv \varphi_P(\rho p).$$

They are strictly concave and have respective maximisers p_H^{PD}/r and p_P^{PD}/ρ . Hence π_ρ is strictly concave. At the smallest of the three transformed maximisers every component has a weakly non-negative derivative, while at the largest every component has a weakly non-positive derivative. The unique zero of π'_ρ therefore lies between them.

Because $p_P = \rho p_\rho^*$ and ρ is fixed, $d \log p_P = d \log p_\rho^*$. For the share comparative statics, use $\alpha_F = 1 - \alpha_H - \alpha_P$ and differentiate the first-order condition

$$\alpha_F \varphi'_F(p) + \alpha_H r \varphi'_H(rp) + \alpha_P \rho \varphi'_P(\rho p) = 0.$$

Holding α_P fixed gives

$$\frac{\partial p_\rho^*}{\partial \alpha_H} = - \frac{r \varphi'_H(rp_\rho^*) - \varphi'_F(p_\rho^*)}{\pi''_\rho(p_\rho^*)}.$$

Holding α_H fixed and varying α_P yields the stated analogous formula. The denominator is strictly negative in both expressions. Under the additional ordering in the proposition, the bracketing result gives $p_F^{PD} < p_\rho^* < p_H^{PD}/r$. Strict concavity therefore implies $\varphi'_F(p_\rho^*) < 0$ and $\varphi'_H(rp_\rho^*) > 0$. The numerator in the displayed comparative static is strictly positive, so $\partial p_\rho^*/\partial \alpha_H > 0$. $\square$